

Shipping Industry and Japanese Shipping Company's Stock Analysis

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September 2026

1 Research topic and background

1.1 Overview of topic

This assignment focused on the shipping industry by surveying related relevant economic and industry data by demonstrating visualized output, such as tables, graphs and maps. In addition, paying attention to Japanese shipping companies to acquire solid understandings of their stock performances and sensitivity and trying to apply effective prediction methods for future stock prices.

1.2 Objectives and motivation

The island countries, such as the United Kingdom and Japan, are disconnected with the continents and citizens in these countries have been depending on oversea trading through the shipping industry to fulfill their shortage of goods. In fact, Japan has few natural resources, such as oil, gas and coal, and has been completely depending on import trading from overseas, in total 755 billion USD in 2025. Import is a lifeline to fulfill the shortage of goods and export is an acquiring method of foreign currencies for the island countries. 99.5% of all import and export in Japan was executed through shipping transportation. To execute the destiny of the countries, shipping companies have essential roles to transport goods for the citizens.

Ships are vital assets for shipping companies but the citizens have enjoyed using a ship, such as boats and yachts for cruising, fishing and marine sports. The author has a small vessel operator's license in Japan and loves cruising too. In addition, the author works as an accounting consultant in Japan and be interested in an analysis of stock performances in a particular industry by using a method of financial modeling. These personal emotions accelerated to start a research for the shipping industry.

1.3 Research questions

To develop an overview of the shipping industry, an author set questions following research questions from macro economical in the world to Japanese micro-level players:

- How about import and export trends over the world and Japan, and which area shows presence.
- How many ships are operated over the world. Whether the number of ships increased or not, and which countries have influences for the ownership of ships.
- Which ports were useful and appreciated for ship trading.
- Which companies has operated over the world. How about the Japanese shipping companies' stock prices and volatility. In the future, what companies stock price would be soared.

1.4 Key findings

China has demonstrated a presence on maritime strategy, such as numbers of ships, beneficial ownerships and port usability, linking with economic strategy. On the other hand, the US and Japan have not reached yet.

Ship transportation was important for Japan, major three ship companies showed strong growth on its stocks. Nippon Yusen and Mitsui OSK demonstrated similar movements and future prediction but Kawasaki Kisen showed high volatility and stayed lower level of the future prediction.

2 Data Sources

- Merchandise total trade: International trade in goods, from UNCTAD. Available at: <https://unctadstat.unctad.org/datacentre/dataviewer/US.TradeMerchTotal> (Accessed: 21st August 2026) Total merchandise exports and imports, expressed in millions of USD by countries and regions.
- Merchant fleet: World fleet by flag of registration and ship type, from UNCTAD. Available at: <https://unctadstat.unctad.org/datacentre/dataviewer/US.MerchantFleet> (Accessed: 26th August 2026) Data are presented in thousands of dead-weight tons (DWT) and the numbers of total fleet with covering 100 gross tons and the above.
- Merchant fleet: World fleet by beneficial ownership - Core indicators - Annual, from UNCTAD. Available at: <https://unctadstat.unctad.org/datacentre/dataviewer/US.FleetBeneficialOwners> (Accessed: 20th August 2026) Number of the sea-going merchant vessels of 1000 gross tons and above. Military vessels, yachts, fishing vessels are excluded.
- Liner shipping: Port connectivity index - Quarterly, from UNCTAD. Available at: <https://unctadstat.unctad.org/datacentre/dataviewer/US.PLSCI> (Accessed: 25th August 2026) Measuring a port's integration into global liner shipping networks. A higher value is associated with better connectivity.
- Yahoo finance. Available at a library on Python (Accessed: 8th September 2026).

3 Data Analysis

The analysis was structured from macro-level factors to micro-level factors, combining with historical records and future forecasts using visualization to extract key insights. Summary of an analysis plan was described below:

Table 1: Summary of observations

No.	Names	Object of observations	Key factors
1	Import and export data on global and Japan	To identify the macro trend of import and export volumes. These factors influence numbers of vessels.	Import volume, Export volume
2	Merchant fleet number and capacity	To identify which area and country show the presence of ship transportation	Merchant fleet number, ton-based capability
3	Beneficial ownership	To provide supportive evidence to the presence	Beneficial owner number
4	Accessible port ranking	To identify the country's maritime strategy for ship transportation by using external evaluation	Port connectivity indexed score
5	Stock price for Japanese	To identify the Japanese shipping player's stock performances	Closing price, differences, Beta, future stock price range

3.1 Import and export data on global and Japan

[Pre-processing]

The data showed that how many trade volume has been generated by region and some specific country. The original data contains long period data from 50's and many region's data, so pre-processing was performed by using codes below:

```
1 import_data = pd.read_csv('TradeMerchTotal_Import.csv',
2                             usecols=[0,*range(53,79)],
3                             index_col=0,
4                             names=['Region',*map(str,range(2000,2026))],
5                             skiprows=1,
6                             nrows=11)
```

After pre-processing, both import and export data contains 11 rows and 26 columns and these data were a same format. To visualize more persuasively, these data were merged as 'Import and Export data'.

[Visualization and extract insights]

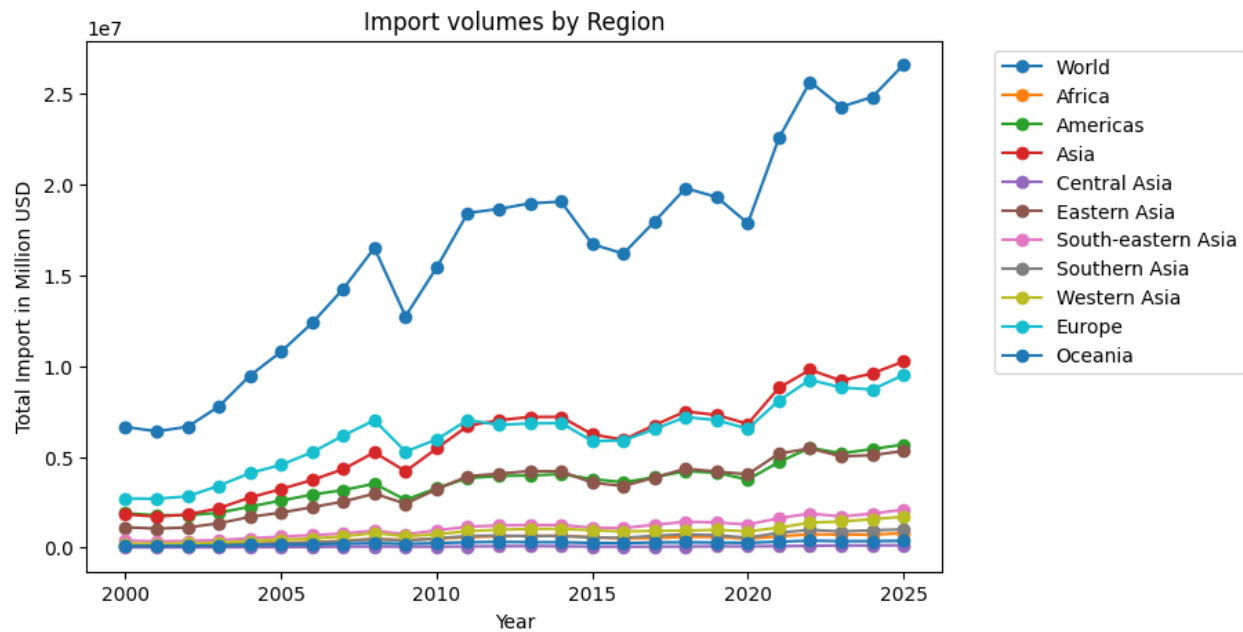


Figure 1: Import volumes by Region

Table 2: China Import and Export in 2020 and 2025 in Million USD

China	2020	2025	Increment
Import	2,065,964	2,582,896	1.25
Export	2,589,952	3,771,842	1.45

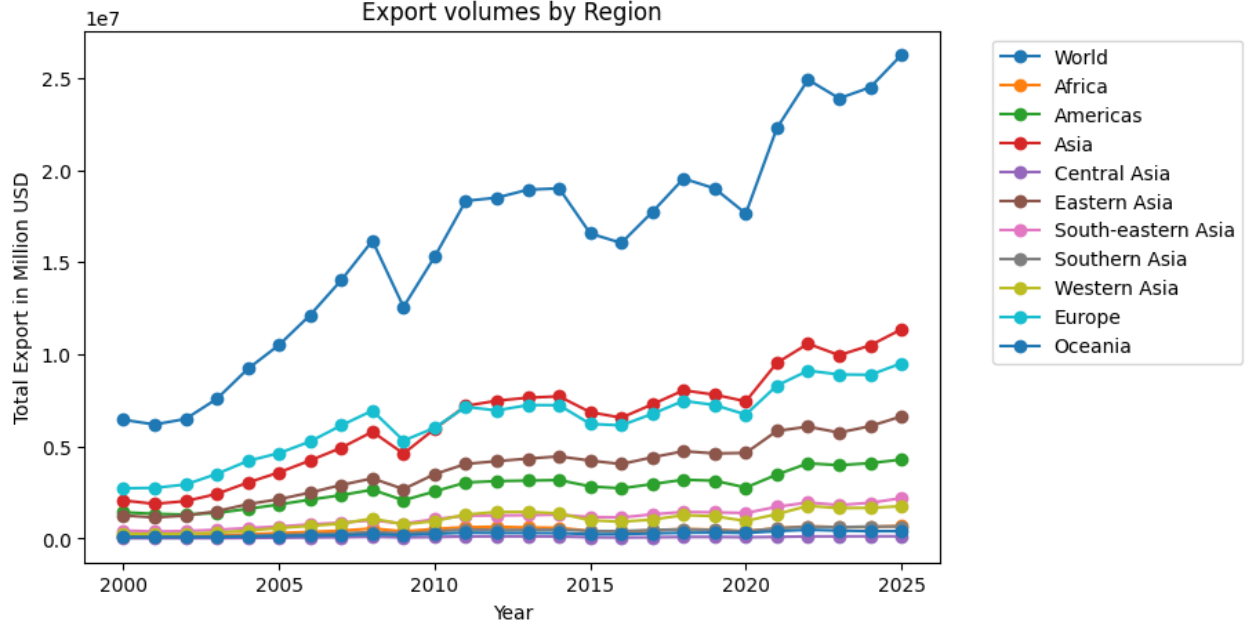


Figure 2: Export volumes by Region

Import and export are inextricably linked and these trends were similar. Even though there were some significant dropping points, such as at the global financial crisis in 2008 and COVID-19, these long-term trends were increasing. People around the world needed goods more and more. Especially, Asia was the greatest importer and exporter. According to the original data table, China showed strong growth for both import and export, and their import accounted for 25.1% and their export accounted for 33.2%, therefore, the China was an growth engine of Asia growth.

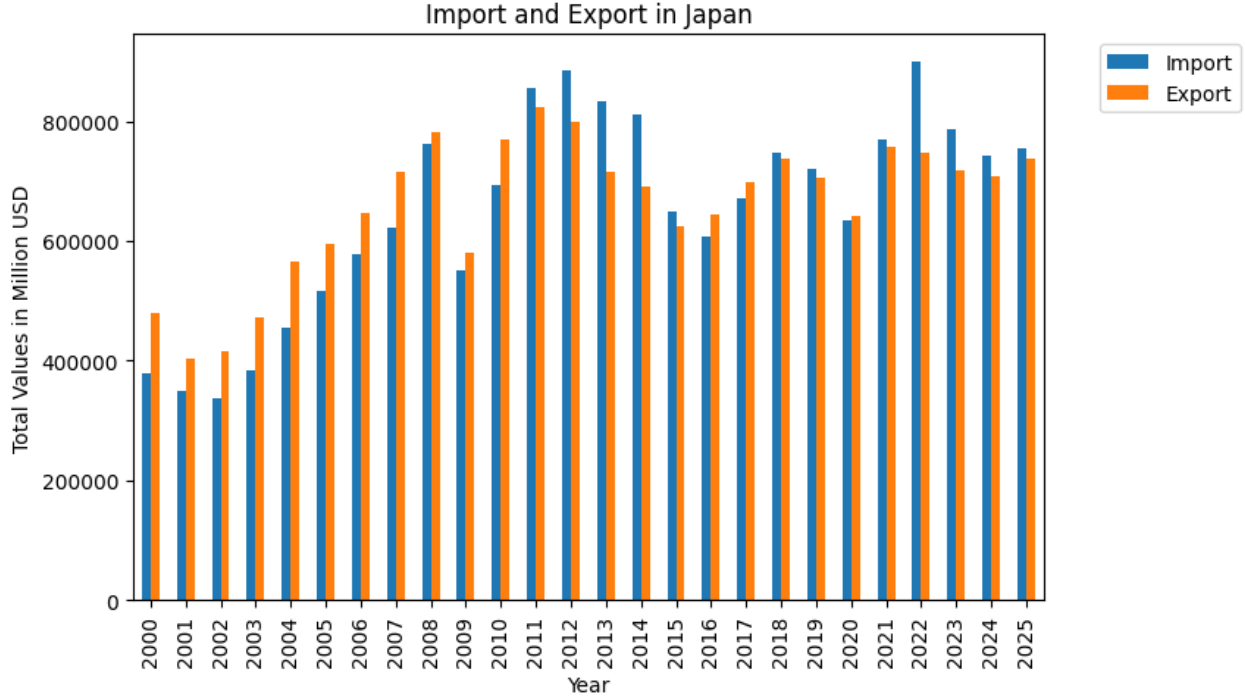


Figure 3: Import and Export in Japan

Focusing on Japan, it was similar movements with the global trend. In recent 5 year from 2020 to 2025, imports exceeded exports, on the other hand, imports were below exports before 2010. Nowadays, Japanese people needed more goods from overseas.

Both world and Japan trends has been uptrend and more transportation capacities have been needed for satisfying demands of people.

3.2 Merchant fleet number and capacity

[Pre-processing]

The data described the number of fleet and transportation capacity by region. The data also contained old data from 80's. To grasp the middle term trends, 15 years, of the fleet number, preprocessed by using 'usecols'. On the other hand, capacities were depicted long term trends without columns selection years, therefore, no used 'usecols' and gathered from 1980.

```
1 fleet_number = pd.read_csv('MerchantFleet_number.csv',
2                             usecols=[0,*range(32,48)],
3                             index_col=0,
4                             names=['Region', *map(str, range(2011,2027))],
5                             skiprows=1,
6                             nrows=6)
```

```
1 fleet_ton = pd.read_csv('MerchantFleet_ton.csv',
2                          index_col=0,
3                          names=['Region', *map(str, range(1980,2027))],
4                          skiprows=1,
5                          nrows=6)
```

[Visualization and extract insights]

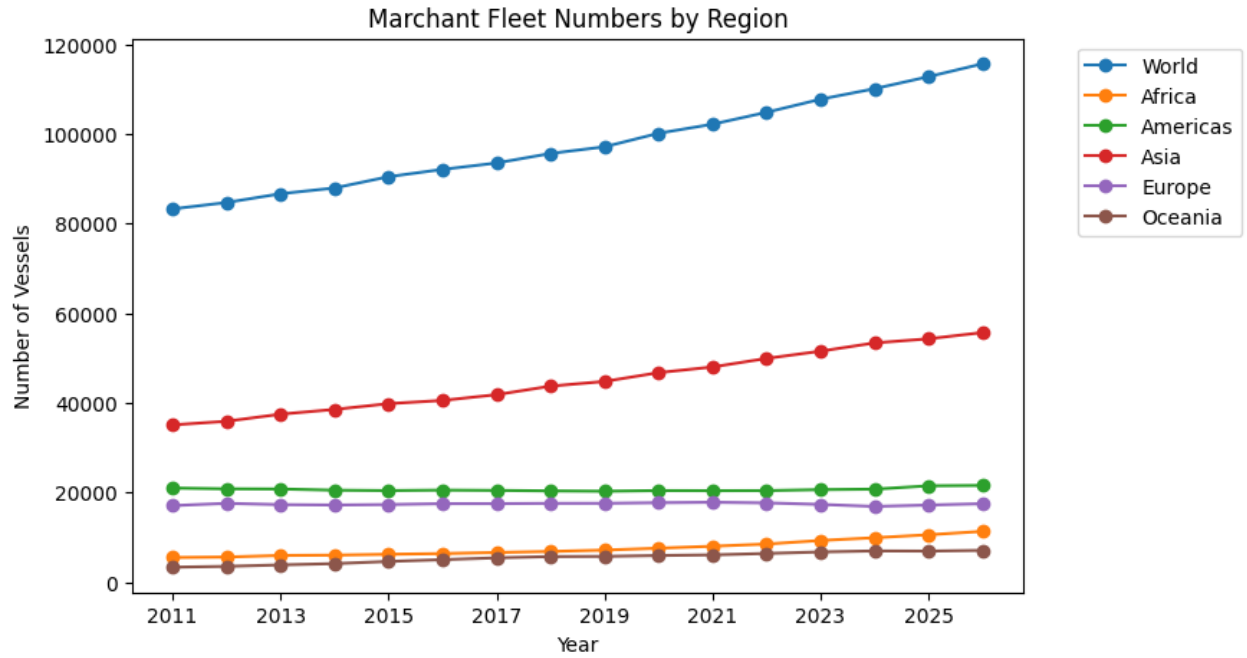


Figure 4: Marchant Fleet Numbers by Region

The number of fleets has been increased in the world. The main growth engines were Asia and Africa. The most part of these areas were development countries and China. The original data does not contain individual company's data, therefore not sure whether China contributed increment or not.

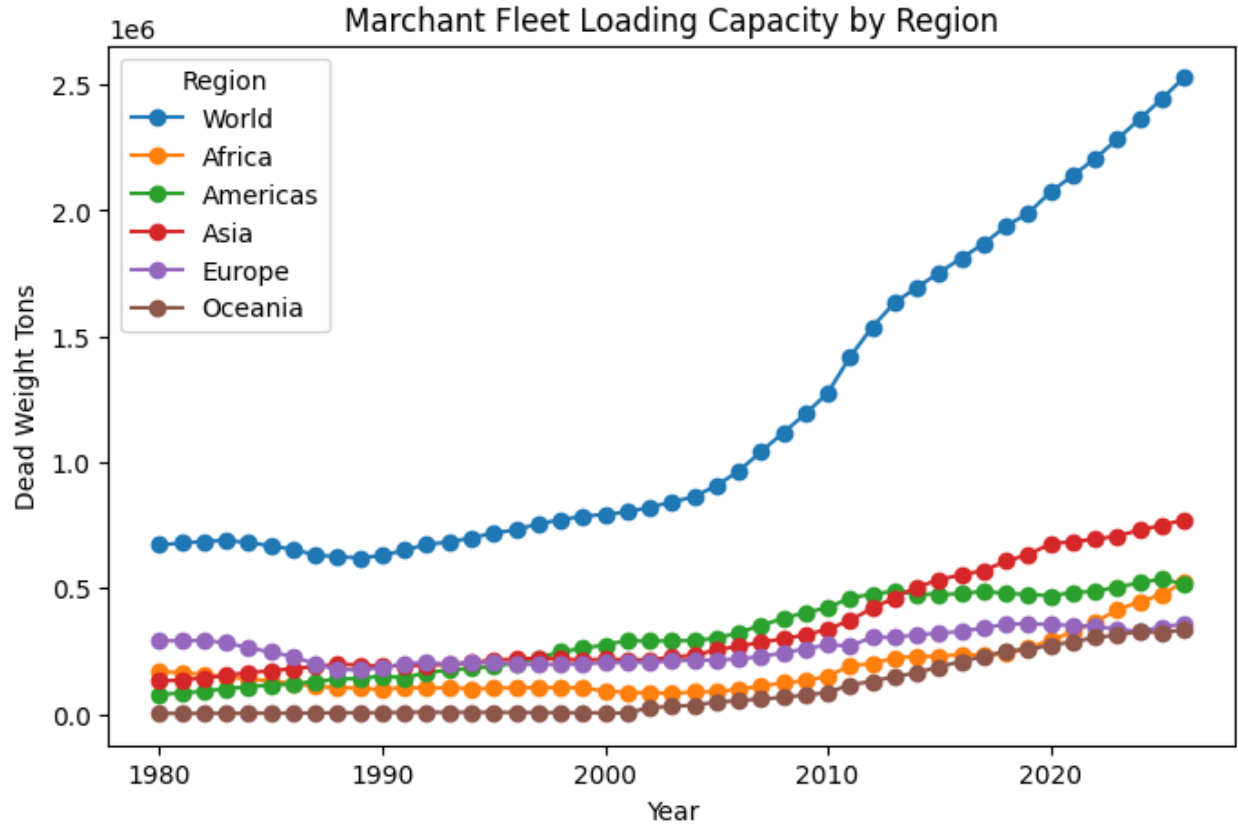


Figure 5: Enter Caption

Long-term trend in transportation capacity showed significant growth during the middle of 2000's to the beginning of 2010's. Transportation capacity also same movements with the number of fleets, Asia and Africa has been increased than the others. In addition, difference between Asia and Africa were relatively small rather than the fleet numbers. This insight showed new fleets in Asia were small capacity or vice versa.

3.3 Beneficial ownership

[Pre-processing]

The data described the number of fleet by beneficial ownership. This data contained the number of fleet in rows and the beneficial ownership country and its aggregate values in columns. To compare countries without the aggregate total, specify the country columns by using 'usecols'. In addition, to visualize smoothly, transposed matrix data by using '.T' and its beneficial ownership name recorded as the country name adding explanatory words such as 'Japan_Number_of_ships_Value'. To extract country name only, preprocessed to delete useless words by 'str.split('_').str[0]'.

```

1 owner_data_26 = pd.read_csv('FleetBeneficialOwners_2026.csv',
2                             usecols=[*range(0,28)],
3                             index_col=0,
4                             nrows=1).T
5 owner_data_26.index = owner_data_26.index.str.split('_').str[0]

```

[Visualization and extract insights]

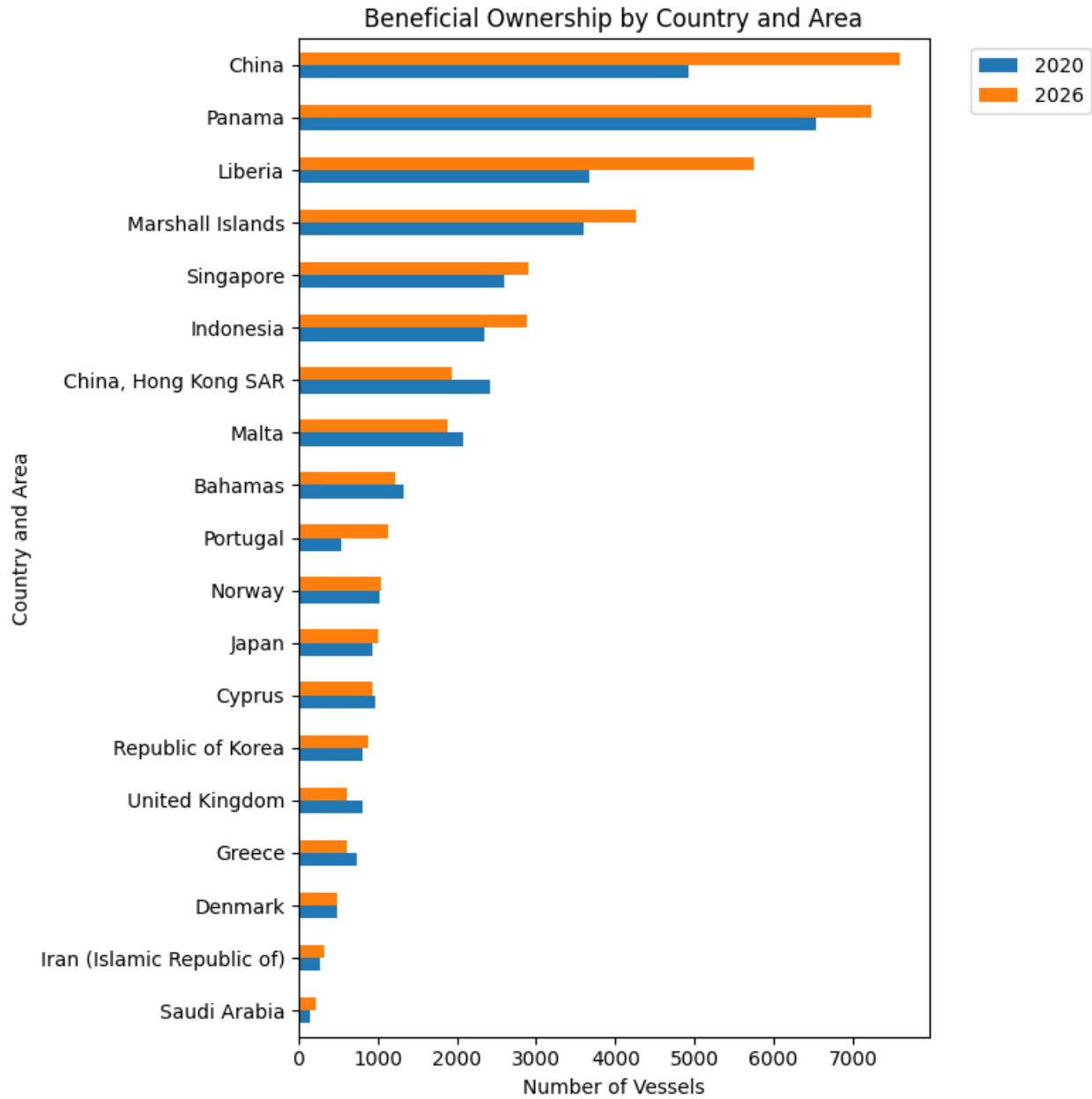


Figure 6: Beneficial Ownership by Country and Area

As per previous research in trading volume and number of fleets, China owned the largest number of big-sized fleets but Panama in Americas and Liberia in Africa showed its presence. Especially Liberia was categorized as development country but they were 3rd rank with significant growth in fleet economy. The reason was flag of convenience, which stated ships must be registered with a single country, even though they operate in international waters, and the country is legally responsible for ensuring that shipowners meet certain basic standards on safety and crew welfare. Both countries eased standards, tax benefits and allowed cheaper foreign labor to operate as a crew, therefore, shipowners choose these countries for their economical merits.

3.4 Accessible port ranking

[Pre-processing]

The data contained port usability index data. This data included 1354 port label in rows and index data in columns from Q1 2006 to Q2 2026. To depict the most recent score in a map, used a single point of data at Q2 2026. And the port label was needed to modify because country and port name were stated in a single column, such as 'Viet Nam, Chan May'. To specify country and port name distinctively, used '.str.split(',')', n=1, expand=True)', to separate data into two columns.

```
1 accessive_port = pd.read_csv('PLSCI_index.csv',
2                             usecols=[0,82],
3                             names=['Port','PLSCI_2026Q2'],
4                             skiprows=1)
5 accessive_port[['Country','PortName']] = accessive_port['Port'].str.split(',',
6                                                         n=1,expand=True)
```

In addition, to depict score data on to a map, latitude and longitude data on each port are needed. 'geocoders', which is an library in python and is used as charge-free.

```
1 from geopy.geocoders import Nominatim
```

In this research, tried to use 'geocoders' but the targeted data was not gathered sufficiently because 'geocoder' equip a significant restriction that a response should complete within a second to manage large amount of accesses, and if it is not completed within the time, an error message appeared such as '(Caused by ReadTimeoutError("HTTPConnectionPool(host='nominatim.openstreetmap.org', port=443): Read timed out. (read timeout=1)"))'. To avoid the critical suspension, extracted latitude and longitude data from Gemini as an alternative approach.

[Visualization and extract insights]



Figure 7: Port Accessibility Top 50 in 2026 2Q

Many port in Asia and Europe are depicted in the map, especially China has many accessible ports, such as

Shanghai, Nigbo and Qingdao. According to the table below, only 3 ports other than Asia were ranked in top 20, Rotterdam, Antwerp and Hamburg. On the other hand, even though the United States is the largest GDP country, only three ports were ranked in top 50, New York/New Jersey, Savannah and Norfolk, from the east coast.

Port accessibility does not directly reflect the economical scale, but China has the strong presence in the port management.

Table 3: Port Accessibility Top 20 in 2026 2Q

Rank	Country	PortName	PLSCI_2026Q2
1	China	Shanghai	2372.47
2	China	Ningbo	2040.69
3	Singapore	Singapore	1833.94
4	Republic of Korea	Pusan	1655.17
5	China	Qingdao	1389.05
6	China	Shekou	1157.85
7	Malaysia	Port Klang	1075.80
8	China	Nansha	1071.13
9	China	Hong Kong SAR, Hong Kong	976.84
10	Netherlands (Kingdom of the)	Rotterdam	950.83
11	Belgium	Antwerp	902.03
12	China	Xiamen	900.05
13	China	Taiwan Province of, Kaohsiung	821.61
14	China	Yantian	795.38
15	China	Xingang	767.16
16	Thailand	Laem Chabang	743.29
17	Sri Lanka	Colombo	719.11
18	Japan	Yokohama	698.98
19	Viet Nam	Haiphong	690.29
20	Germany	Hamburg	690.06

3.5 Stock price for Japanese shipping company, volatility and forecast

[Pre-processing]

Yahoo finance provides wide range of stock and accounting data to investors as a charge-free, not only US listed companies but also listed companies in Japanese market. To extract major shipping companies' stock closing data, specified its ticker and downloaded from yahoo finance.

```

1 tickers = ["9101.T", "9104.T", "9107.T"]
2 stock_data = yf.download(tickers, period="5y")
3 st_close = stock_data['Close'].copy()
4 st_close.columns = ['Nippon Yusen', 'Mitsui OSK', 'Kawasaki Kisen']

```

In addition, to identify the cross-variable relationships with each stock closing prices, prepared stock price difference data from previous day by using 'shift()'. Difference data is treated as stationary data, which has features that its average is 0 and its deviation is constant, and is useful for time series analysis. The concept that using one day ago data from each company in the same industry was introduced from VAR(1) model. VAR is abbreviated from 'Vector Auto Regression' and reflected own and other stock prices one day ago to today's own stock price.

```

1 yusen = yf.Ticker('9101.T').history(start='2019-01-01', interval='1d')['Close']
2 shosen = yf.Ticker('9104.T').history(start='2019-01-01', interval='1d')['Close']

```

Table 4: Market capitalization of global shipping companies, 2025(thousand million USD)

Rank	Company	Country	Market Capitalization
1	A. P. Moller-Maersk	Denmark	269.5
2	Hapag-Lloyd	Germany	265.0
3	COSCO SHIPPING Holdings	China	251.1
4	Evergreen Marine	Taiwan	143.8
5	Nippon Yusen	Japan	142.3
6	Mitsui OSK	Japan	120.9

```

3 kisen = yf.Ticker('9107.T').history(start='2019-01-01', interval='1d')['Close']
4
5 yusen_diffs = (yusen - yusen.shift())[1:]
6 shosen_diffs = (shosen - shosen.shift())[1:]
7 kisen_diffs = (kisen - kisen.shift())[1:]

```

And to demonstrate interdependency between each company, used Beta as following equation.

$$\beta_i = \frac{\text{Covariance}(\text{Return}_i, \text{Return}_m)}{\text{Variance}(\text{Return}_m)} \quad (1)$$

Beta demonstrated how degree of price influence to others. If the output is over 1, it shows a target company would be more volatile from other company. When other company's stock price increased 5%, a target company's stock price increased more than 5%. In a decline situation, a target company's stock price shows similar large dropping.

[Visualization and extract insights]

Before diving deeper into analyzing Japanese shipping companies, demonstrated global players:

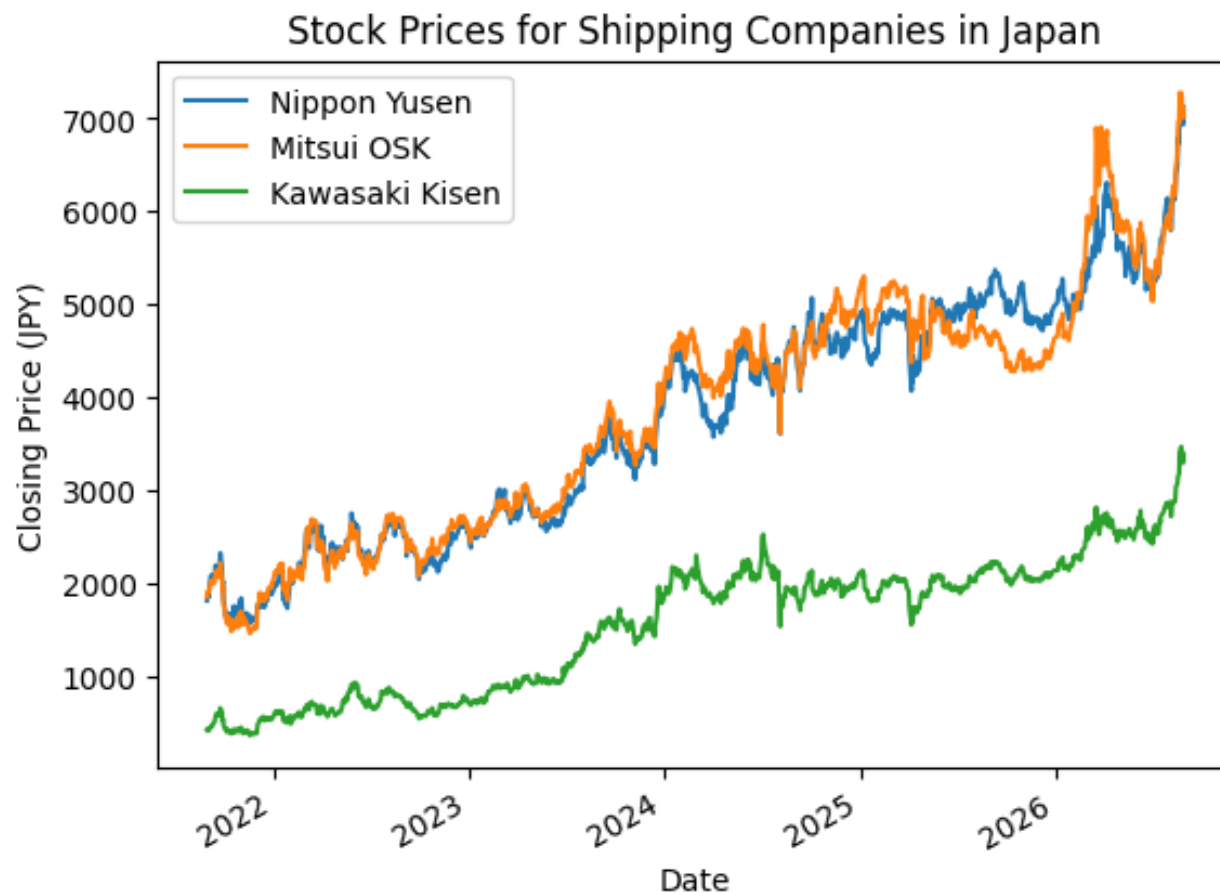


Figure 8: Stock Prices for Shipping Companies in Japan

In general, the stock price of three Japanese shipping companies have been similar growth trend from 2022 to 2026. In detail, the ranges of decline were different at the first half of 2024, Mitsui OSK recorded a significant decrease, and the second half of 2025, Nippon Yusen dropped. As an another difference appeared at the beginning of 2026, Nippon Yusen and Mitsui OSK recorded significant increase but Kawasaki Kisen had been steady.

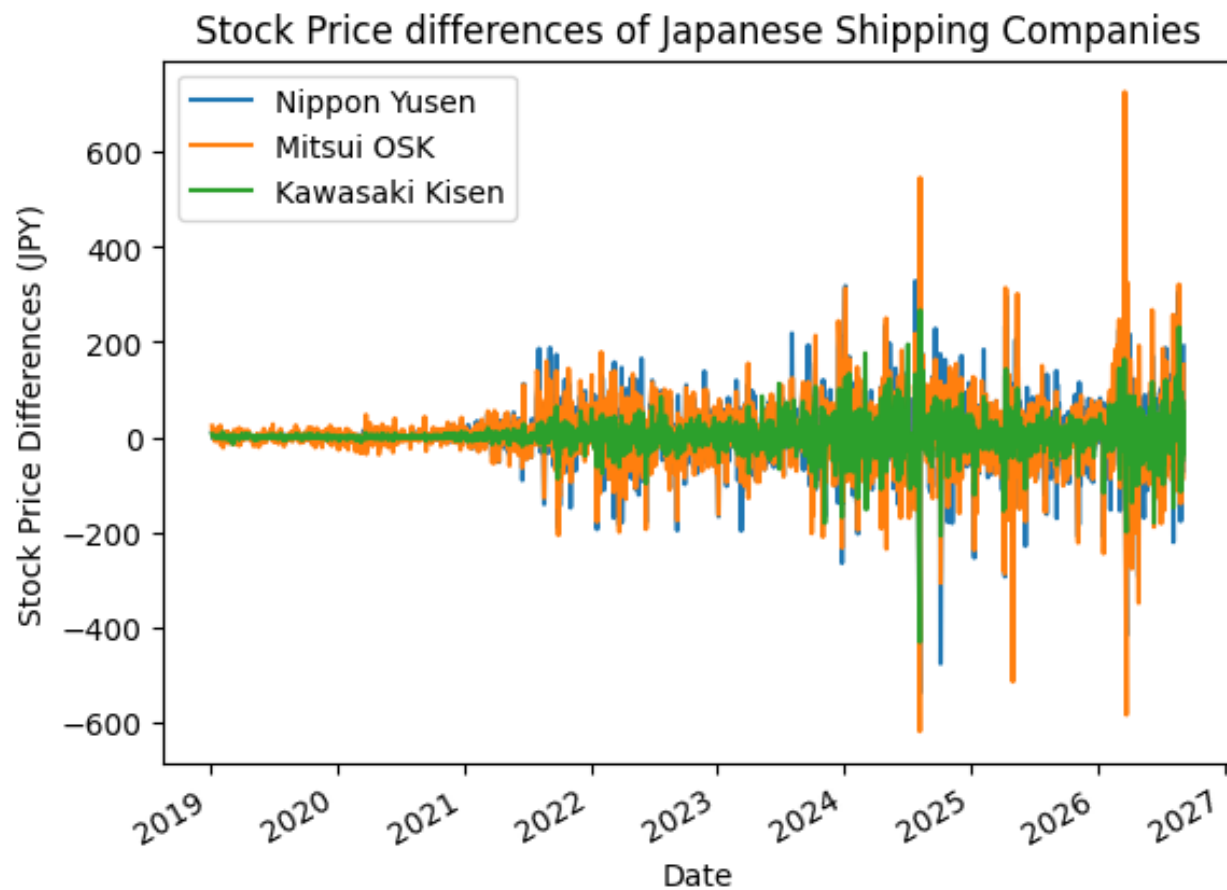


Figure 9: Stock Price Differences of Japanese Shipping Companies

In the view of differences of stock prices, points of spikes were emphasized rather than the closing prices. Mitsui OSK recored significant spikes at the second half of 2024, the first half of 2025 and the first half of 2026. Kawasaki Kisen recorded significant spike with Mitsui OSK at the second half of 2024. On the other hand, Nippon Yusen recorded less volatile differences compared other two companies.

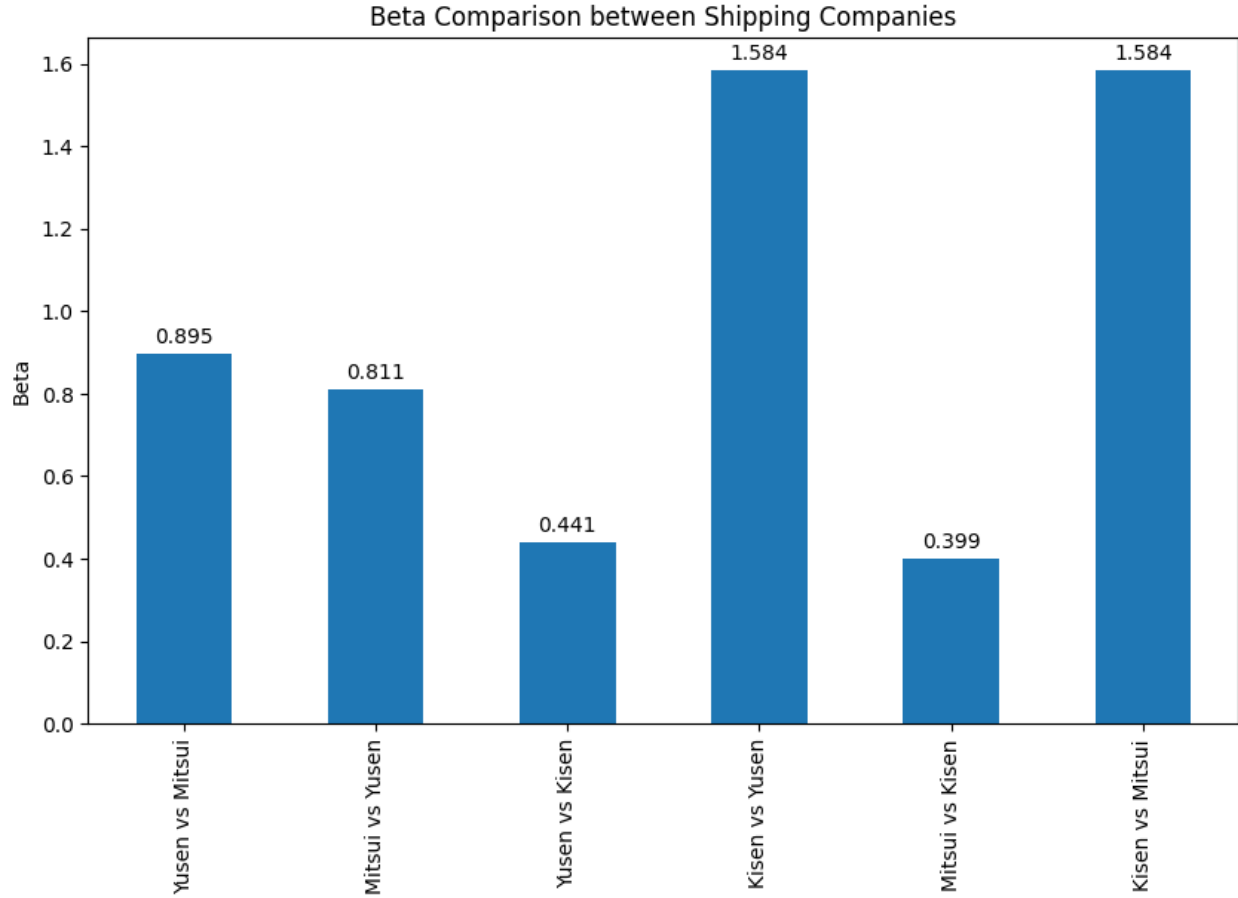


Figure 10: Beta Comparison between Shipping Companies

The Beta demonstrated sensitivity of stock price each other. Beta between Nippon Yusen and Mitsui OSK are 0.8 to 0.9 and their stock prices are synchronized on the whole. On the other hand, Kawasaki Kisen illustrated significant volatility toward the two companies, coincidentally both scores were 1.584, even though the two companies have not been influenced from Kawasaki Kisen, 0.441 and 0.399. The results suggested that Nippon Yusen and Mitsui OSK have similar portfolio and performances but Kawasaki Kisen has focused other types of business and specialized different features.

95% of prediction interval on three companies showed that Nippon Yusen and Mitsui OSK are plotted similar price ranges but Kawasaki Kisen is still lower range.

4 Conclusion

4.1 Summary of key findings

Through the processes of the research, identified that China has presences on import, export, ship ownerships and port usability. Not only expanding trade activities but also they have owned a lot of ships and developed convenient ports around coastal areas. Linked with a maritime strategy and an economic strategy. In contrast, the US, the largest GDP country, was not demonstrating a presence in maritime fields. Interestingly, Panama and Liberia were chosen to register ships by easing regulation and taxation.

In Japan, import and export have grown after the pandemic in 2019. Even though Japan has economic volume and its marine transport is essential, they do not show a presence like China. The rank of the beneficial

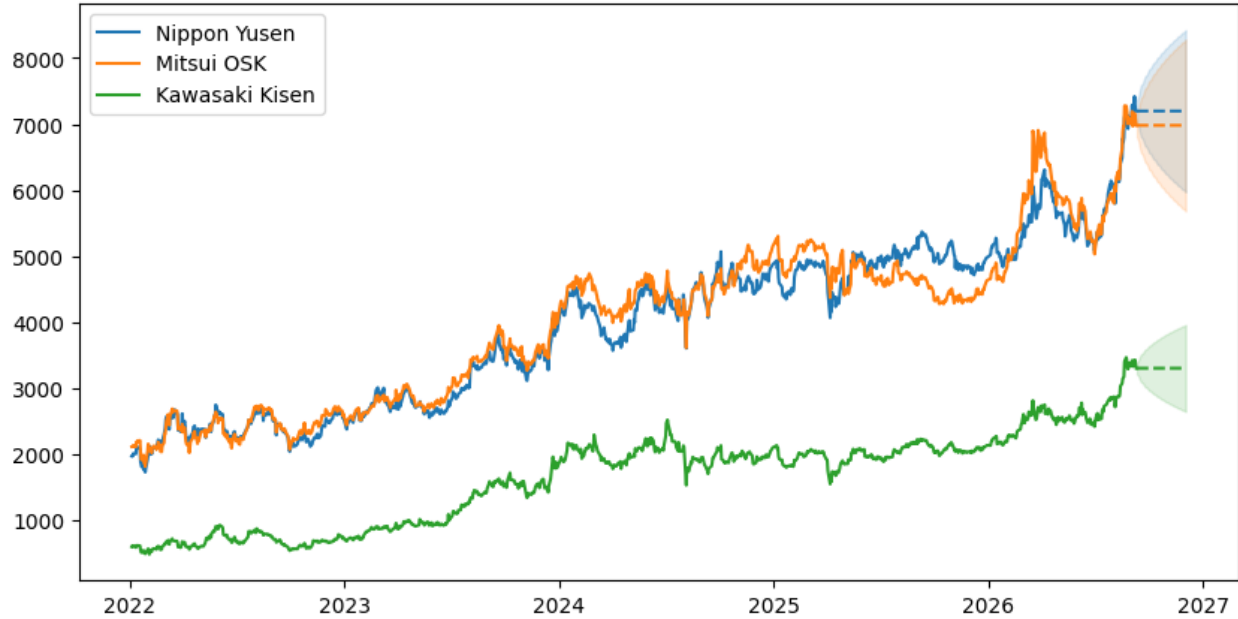


Figure 11: Stock Price Forecasts Shipping Companies

ownership was 12th and there was a port, Yokohama port, in the rank of port usability top 20. On the other hand, shipping industry in Japan has been growth and major three shipping operators, Nippon Yusen, Mitsui OSK and Kawasaki Kisen, recorded historical high stock prices. Their stock showed similar movements but Kawasaki Kisen has high volatile compared with other two companies.

4.2 Evaluation of the research

Main achievements of the research were completed a whole analysis process in Python, which gathering essential data from reliable sources to demonstrate visualized outputs, and described the overview of ship industry in both macro economical view and micro economical view. It is possible the result of the research to use as an underlying study and extend further research in the maritime field.

4.3 Challenges and limitations

As challenges to expand own capability, the author tried to contain newly learned knowledge and skills, such as pre-processing methods, time-series analysis and geospatial visualization in researching processes. These methods contributes to not only research capability but also work capability in consulting fields.

Even though the knowledge of the shipping industry was not enough, described the brief overview of the industry and its trends. At the same time, an author felt a limitation, which was the depth of questions were not deep and the coverage of data was narrow. When a planning of a research targeted to a deeply innovate theme, it is better to target matters that a researcher has enough domain knowledge or repeated researches in the similar fields.

4.4 Problems for future research

To set critical questions for finding innovative research results, need to watch the industry in narrow view on the shipping industry rather than generalized broad view, with combining the knowledge and skill on areas, which a researcher has enough domain capability.

5 Critical Engagement with AI

The object was to demonstrate the overview of the shipping industry by using learned knowledge and technics in the course of Data Visualization. Now a days, Generative AI can walk through each process, such as data gathering, depicting graphs and deriving insights, for covering all over a research alternatively. However, heavy dependence on Generative AI does not lead to strengthening the research capability and they equip inherent risks, such as hallucination, data leak and copyright infringement. This is a limitation of AI. A researcher should control AI risks properly and commit in the contents of the final output on own responsibility.

In this research, an author used Generative AI as following usages:

1. An author does not have enough knowledge about the shipping industry. Gemini (<https://gemini.google.com/app?hl=ja>) taught me an authoritative book, Maritime Economics(1997), to understand the industry. Entered at 16th August 2026: "I am going to write a work paper as an assignment of the Data Visualization course in Master of Degree program in Data Science. I focused on the shipping industry but I do not have enough knowledge about it. Tell me an authoritative paper or book to understand the industry" in Japanese.
2. Gemini (<https://gemini.google.com/app?hl=ja>) generated port's latitude and longitude data instead of using a python library 'geocoders' due to limitation of data extraction. Entered the late of August 2026: "Used by inserted port data, tell me latitude and longitude data for visualizing scores on the map generated pydeck." in Japanese.
3. The research output was made by using Latex in Overleaf. An author does not have enough skill to use it but believed the tool is useful for writing academic paper in the future doctoral program. Gemini (<https://gemini.google.com/app?hl=ja>) supported and instructed to use Overleaf. Entered the each time during assignment period from 10th August 2026 to 14th September 2026: example prompts were "Tell me how to use Overleaf. I tried to copy python code on Overleaf but the edge on the code out of PDF format. How to solve this issue " and "Tell me how to insert a graph made in JupyterNotebook to Overleaf" in Japanese.

Avoided to use Generative AI in the phase of determining object and a later phase, such as extracting insights from the result of data visualization and making conclusion, because setting theme is depending on an author's personality and the responsibility of the final output belongs to an author. Tried to insert personal story in brief to explain the motivation of the research.

6 References

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7 Word Count

This report contains 2817 words excluding references.